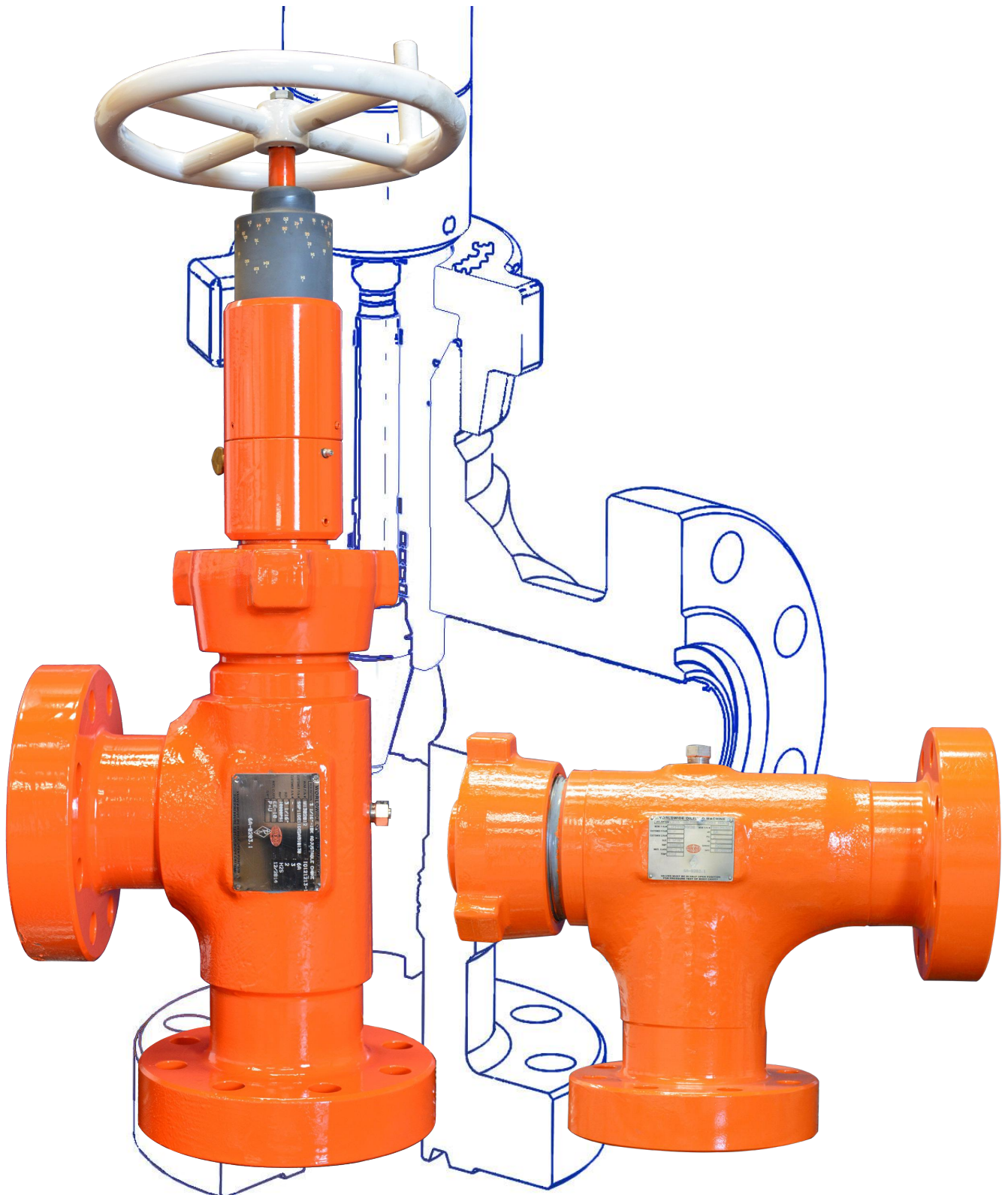




WORLDWIDE OILFIELD MACHINE

www.womusa.com

Chokes



Invention, Unique Design and Extraordinary Craftsmanship



WOM proudly celebrates 39 years of providing the oil & gas industry with meticulously designed, well-engineered pressure and flow control equipment. WOM looks forward to continuing to make a positive impact on the process of safely extracting one of the world's most precious resources. We are honored to work with the businesses that have trusted our commitment to excellence and appreciate each member of our group that has contributed to our success.

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Choke Intro



TM **WORLDWIDE
OILFIELD
MACHINE**

Utilizing a Worldwide Manufacturing and Engineering presence, WOM Chokes offer affordable and long term solutions for various choke applications.

WOM Chokes are engineered and manufactured in compliance with the standards:

- API 6A / ISO 10423
- PR1 AND PR2
- ANSI B16.5
- NACE Mr0175
- ISO 9001/2008
- EN 10204
- API 16C

Applications and Industries served

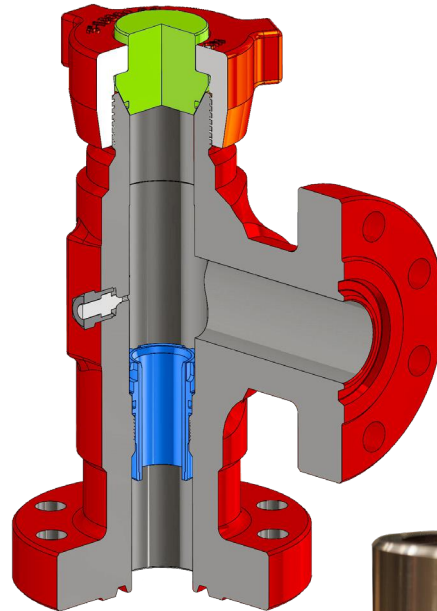
- Production
- Flowback
- Water Injection
- Gas Injection and Gas Lift
- Chemical Injection
- Fracturing Clean Up
- High Pressure Drop Application
- Multi-Phase Flow Application
- Cementing
- Drilling Manifolds
- Gas Processing



Positive and Adjustable Choke



WOM Positive Choke



Flow Bean

WOM Positive Choke

The WOM Positive Choke is a simple design that utilizes an angle body with a blanking plug assembly and a threaded replaceable flow bean.

Available in all standard sizes.

Flow beans have a smooth entrance to provide a smooth flow while allowing minimal turbulence. Beans are available in both integral and fractional 64th sizes.

WOM Adjustable Choke

The WOM Adjustable, Needle and Seat Choke, provides a simple and economic solution to multiple applications, including drilling manifolds and production trees and many more. The simple design consist of an angled choke body with an adjustable bonnet assembly and seat. A rising and rotating stem, operated by a handwheel, allows ease of operation. As an added feature, the adjustable choke can easily be converted to an external sleeve design.

Major parts of assembly include:

- Conical stem tip (Rising needle)
- Stainless steel seat and indicator (1/64" increments)
- Tungsten Carbide tipped stem and seat available for high erosive services.

Available in all standard sizes.



WOM Adjustable Choke

Features and Benefits

Features and Benefits

- Interchangeable components provide a maximum flexibility of choice.
- Available in both manual and automated operation.
- Meet or exceed API 6A requirements.

Available Sizes and Pressure Ratings

Adjustable and Positive Choke Sizes

2" Nominal Body (1" Max orifice)						
Size / Pressure	2 K	3K	5K	10K	15K	20K
2 1/16"		X	x	x	x	x
2 9/16"			x	x	x	x
3" Nominal Body (2" Max orifice)						
Size / Pressure	2 K	3K	5K	10K	15K	20K
2 9/16"			x	x	x	x
3 1/8"		x	x			
3 1/16"				x	x	x
4 1/16"			x	x	x	
4" Nominal Body (3" Max orifice)						
Size / Pressure	2 K	3K	5K	10K	15K	20K
4 1/16"		x	x	x	x	
5 1/18"			x	x		

Specifications
Nominal Size 2" to 4"
Working Pressure Up to 20,000 PSI *May vary according to size
Temperature: -20 °F to 350 °F
PSL: 1-3
Material Class BB to HH
PR1

WFB, WOM Flowback Chokes

The WOM Flowback choke, Model WFP is designed with Hammer Union connections and is well suited for fracking, flow back, and well service applications.

Available in both a 2" body with a 1" maximum orifice and a 3" nominal body with a maximum orifice size of 2".



Adjustable Choke Specifications

End Type	Style	Connection Type	Temp	Service Type	Dim. A {in / mm}	Dim. B {in / mm}
Model : WFB-20 2" 1502F x 2" 1502M	0.75 Seat	Hammer Union	Low	Std.	9.91 / 251.7	6.06 / 153.9
Model : WFB-20 2" 1502F x 2" 1502M	0.75 Seat	Hammer Union	Low	H2S	9.91 / 251.7	6.06 / 153.9
Model : WFB-20 2" 1502M x 2" 1502F	0.75 Seat	Hammer Union	Low	Std.	9.91 / 251.7	6.06 / 153.9
Model : WFB-20 2" 1502F x 2" 1502F	0.75 Seat	Hammer Union	Low	Std.	9.91 / 251.7	6.06 / 153.9
Model : WFB-20 2" 1502F x 2" 1502M	0.75 Seat	Hammer Union	Low	Std.	9.91 / 251.7	6.06 / 153.9
Model : WFB-20 2" 1502F x 2" 1502M	1.00 Seat	Hammer Union	Low	H2S	9.91 / 251.7	6.06 / 153.9
Model : WFB-20 2" 1502M x 2" 1502M	1.00 Seat	Hammer Union	Low	Std.	9.91 / 251.7	6.06 / 153.9
Model : WFB-20 2" 1502M x 2" 1502F	1.00 Seat	Hammer Union	Low	Std.	9.91 / 251.7	6.06 / 153.9
Model : WFB-30 3" 1502F x 3" 1502M	2.00 Seat	Hammer Union	Low	Std.	11.88 / 301.8	8.38 / 212.9
Model : WFB-30 3" 1502F x 3" 1502M	2.00 Seat	Hammer Union	Low	H2S	11.88 / 301.8	8.38 / 212.9

Control Chokes

The WOM control chokes are available with a plug & cage or an external sleeve trim. These chokes are designed to provide accurate flow control throughout its operating range.

Plug and Cage Choke - WPC Series

Plug and Cage trim is a robust design with maximum flow capacity, making it ideal for oil production and water injection services.

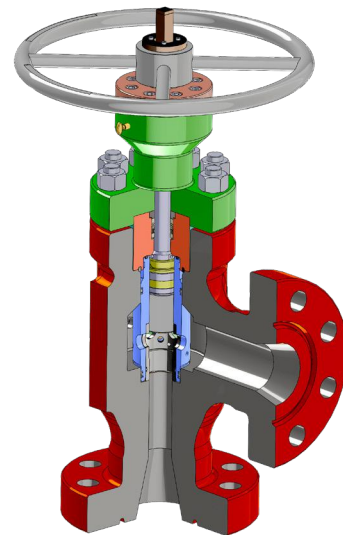
This trim design is used for high capacity/Medium pressure drop application

The WOM Plug and Cage style choke features a tungsten carbide cage as the throttling mechanism with a protective steel carrier around it.

Outer Steel carrier is for protection against impacts from debris in the production fluid.

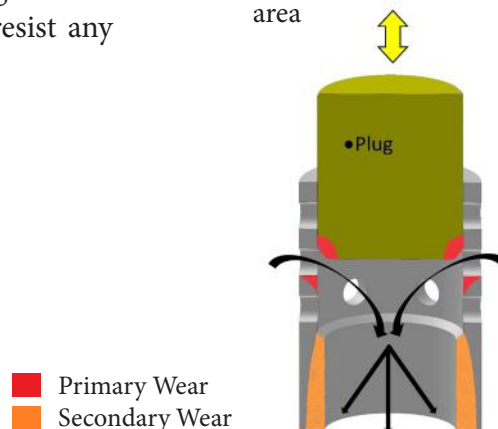
The trim characteristics is equal percentage that provides superior flow control, however, WOM can provide the linear trim as well on demand.

WOM Plug & Cage style design is pressure balanced type so as to minimize the stem loads. Plug is fully guided at the ID and is rigidly attached to the stem to resist any induced vibration damage.



Plug & Cage Trim

Plug moves up/down to vary hole opening area



Control Chokes

Available Sizes and Pressure Ratings

Plug & Cage Choke

CHOKE MODEL	NOMINAL SIZE	AVAILABLE TRIMS SIZES	AVAILABLE TRIMS SIZE S FRACTIONS	WORKING PRESSURE	FLANGE SIZE
WPC20	2"	0.50"	32/64	API 5000 to 15000 PSI ANSI 900# to 2500#	API 1-13/16" to 3-1/16" API 1-13/16" to 2-1/16" ANSI 2" to 3"
		0.75"	48/64		
		1.0"	64/64		
WPC30	3"	0.50"	32/64	API 5000 to 15000 PSI ANSI 900# to 2500#	API 2-9/16" to 5-1/8" API 2-9/16" to 3-1/16" ANSI 2" to 4"
		0.75"	48/64		
		1.0"	64/64		
		1.5"	96/64		
		2.0"	128/64		
WPC40	4"	1.0"	64/64	API 5000 to 10000 PSI ANSI 900# to 2500#	API 3-1/16" to 5-1/8" ANSI 3" to 5"
		1.5"	96/64		
		2.0"	128/64		
		2.5"	160/64		
		3.0"	192/64		
WPC60	6"	3.0"	192/64	API 5000 ANSI 900# to 1500#	4-1/16 to 7-1/16" ANSI 4" to 8"
		4.0"	256/64		
		5.0"	320/64		
		6.0"	384/64		

External Sleeve Choke

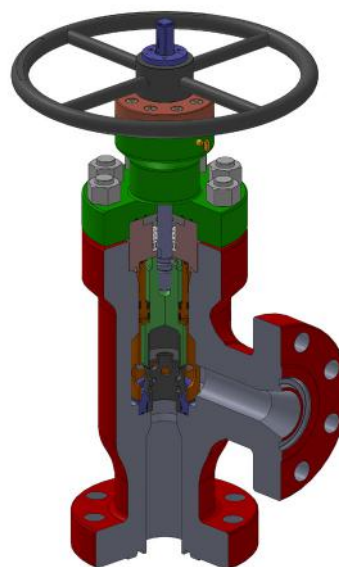
External Sleeve Choke- WES Series

External Sleeve trims possess the same features & are used in high erosive and high pressure drop service. The externally guided flow sleeve controls the opening and rate of flow.

This choke is used for low capacity/high pressure drop applications.

The flow is directed upward away from the outlet, impinges on itself in the centre of the flow cage which in turn isolates the body bore from incoming turbulent flow hence eliminating the body wear.

The seating is achieved through an isolated sealing element contact on a seat outside the flow cage. Keeping the seat surface away from the high velocity flow protects the seat from throttling wear and the seat integrity is maintained.

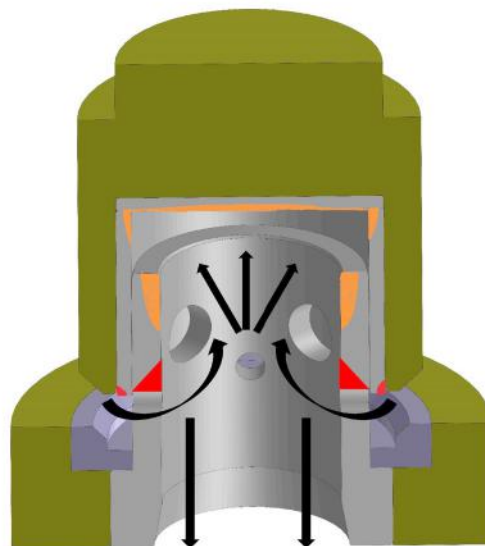


External Sleeve Choke

External Sleeve moves up/down
to vary hole opening area



■ Primary Wear
■ Secondary Wear



External Sleeve Choke

Available Sizes and Pressure Ratings

External Sleeve Choke

CHOKE MODEL	NOMINAL SIZE	AVAILABLE TRIMS SIZES	AVAILABLE TRIMS SIZE FRACTIONS	WORKING PRESSURE	FLANGE SIZE
WES20	2"	0.50"	32/64	API 5000 to 10000 PSI API 15000 to 20000 PSI ANSI 900# to 2500#	API 1-13/16" to 3-1/16" API 1-13/16" to 2-1/16" ANSI 2" to 3"
		0.75"	48/64		
		1.0"	64/64		
WES30	3"	0.50"	32/64	API 5000 to 10000 PSI API 15000 to 20000 PSI ANSI 900# to 2500#	API 2-9/16" to 5-1/8" API 2-9/16" to 3-1/16" ANSI 2" to 4"
		0.75"	48/64		
		1.0"	64/64		
		1.5"	96/64		
		2.0"	128/64		
WES40	4"	1.0"	64/64	API 5000 to 10000 PSI ANSI 900# to 2500#	API 3-1/16" to 5-1/8" ANSI 3" to 5"
		1.5"	96/64		
		2.0"	128/64		
		2.5"	160/64		
		3.0"	192/64		
WES60	6"	3.0"	192/64	API 5000 ANSI 900# to 1500#	4-1/16 to 7-1/16" ANSI 4" to 8"
		4.0"	256/64		
		5.0"	320/64		
		6.0"	384/64		

Standard Materials of Construction

API 6A Material Class and Material						
Item	AA, BB	EE - 0.5 , EE - 1.5	EE - NL	CC, FF -0.5, FF 1.5	FF - NL	HH, HH-NL
Body Bonnet	AISI 4130	AISI 4130	AISI 4130	AISI 410 SS or Grade F6NM SS	AISI 410 SS or Grade F6NM SS	AISI 4130 w/ Alloy 625 Cladding
Stem	AISI 410 or 17 -4 PH SS	AISI 410 or 17 -4 PH SS	INC 718 or INC 725	AISIS 410SS, Grade 17-4 PH SS or GR F6NM SS	INC 718 or INC 725	NC 718 or INC 725
Bolting	ASTM A193 B7 or ASTM A320 L7	ASTM A193 B7 or ASTM A320 L7	ASTM A193 B7 or ASTM A320 L7	ASTM A193 B7 or ASTM A320 L7	ASTM A193 B7 or ASTM A320 L7	ASTM A193 B7 or ASTM A320 L7
Seat, Cage, External Sleeve, Plug	AISI 410 SSS and Tungsten Carbide	AISI 410 SSS and Tungsten Carbide	AISI 410 SSS and Tungsten Carbide	AISI 410 SSS and Tungsten Carbide	AISI 410 SSS and Tungsten Carbide	CRA 718 / 725 / 625 and Tungsten Carbide
Stem Packing	PTFE / Elastomeric	PTFE / Elastomeric	PTFE / Elastomeric	PTFE / Elastomeric	PTFE / Elastomeric	PTFE / Elastomeric

Notes:

1 - H2S concentration as defined by ISO 15156 (NACE MRO175)

2 - H2S no limit as defined by ISO 15156 (NACE MRO175)

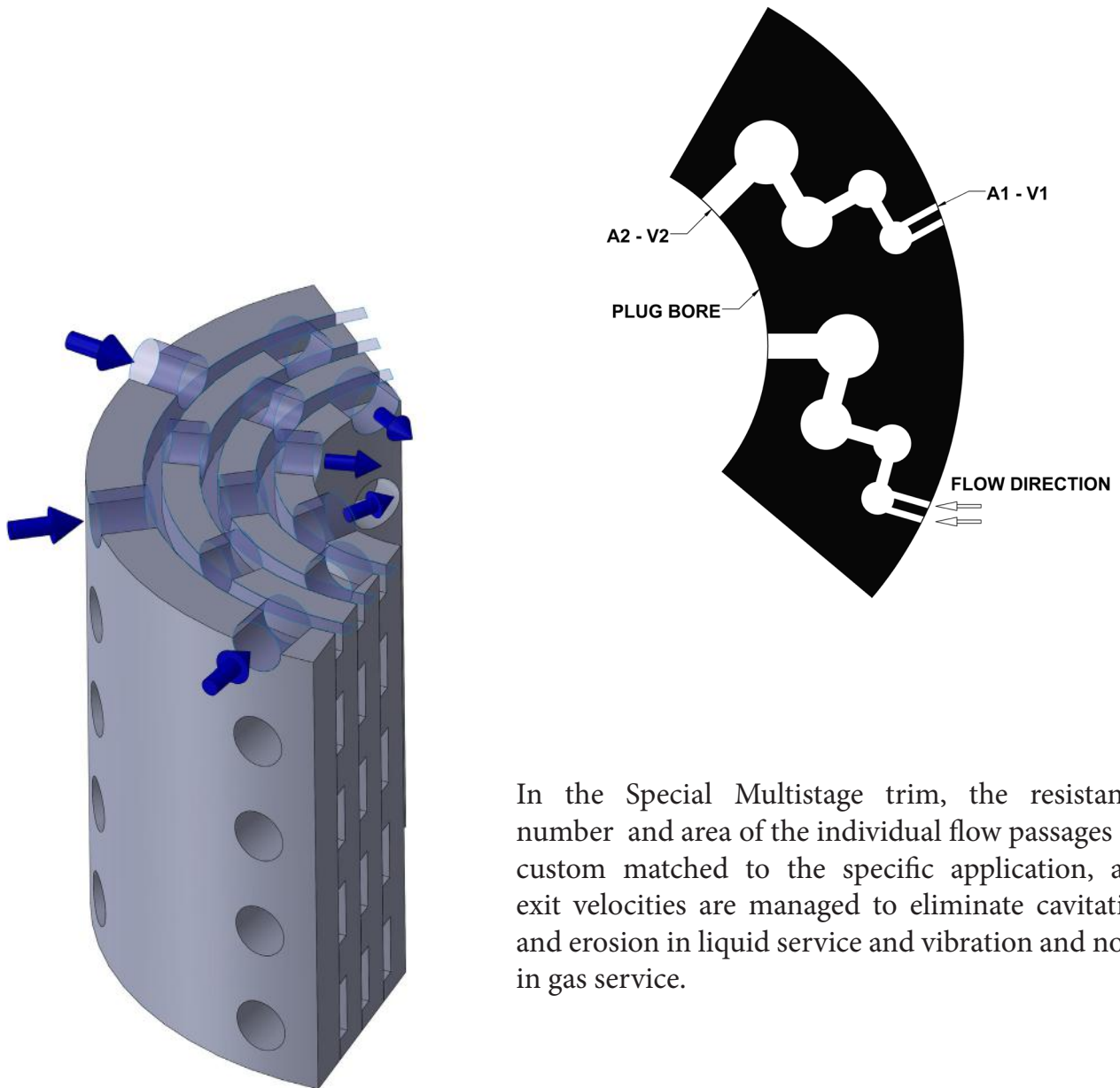
3 - Material selection subject to change and review

Material Class		Minimum Material Requirements	
		Body, Bonnet, Inlet and Outlet Connections	Pressure- Controlling parts and stems
AA	General Service	Carbon or low-alloy steel	Carbon or low-alloy steel
BB	General Service	Carbon or low-alloy steel	Stainless Steel
CC	General Service	Stainless Steel	Stainless Steel
DD	Sour Service	Carbon or low-alloy steel	Carbon or low-alloy steel
EE	Sour Service	Carbon or low-alloy steel	Stainless Steel
FF	Sour Service	Stainless Steel	Stainless Steel
HH	Sour Service	Corrosion-Resistant Alloys (CRA)	Corrosion-Resistant Alloys (CRA)

API Classification	Operating Temperature			
	Min (C)	Max (C)	Max (F)	Min (F)
L	-46	82	-50	180
N	-46	60	-50	140
P	-29	82	-20	180
S	-18	60	0	140
T	-18	82	0	180
U	-18	121	0	250
X	-18	180	0	350

Special Multistage Trim

Special energy dissipating, velocity controlled, disc stack trim can be provided as a solution to severe service applications. Including anti cavitation, high pressure drop, and low noise services.



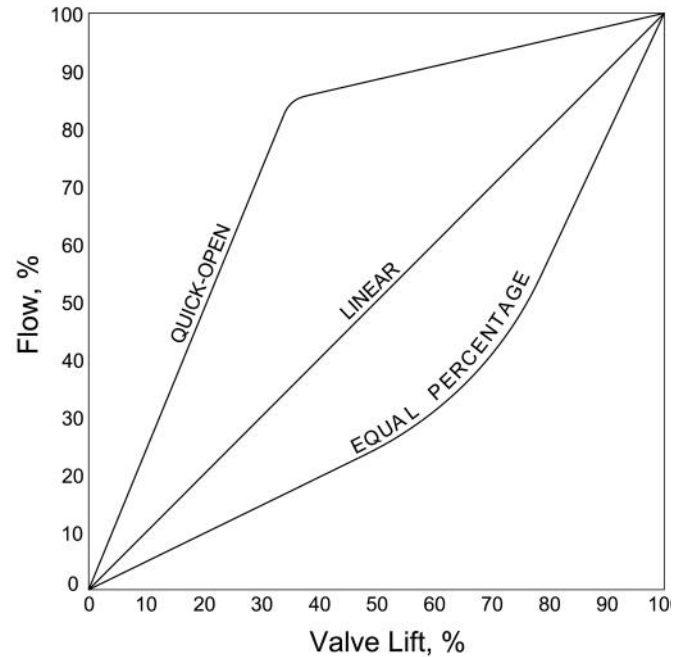
In the Special Multistage trim, the resistance, number and area of the individual flow passages are custom matched to the specific application, and exit velocities are managed to eliminate cavitation and erosion in liquid service and vibration and noise in gas service.

Control Flow Characteristics

Trim Characteristics

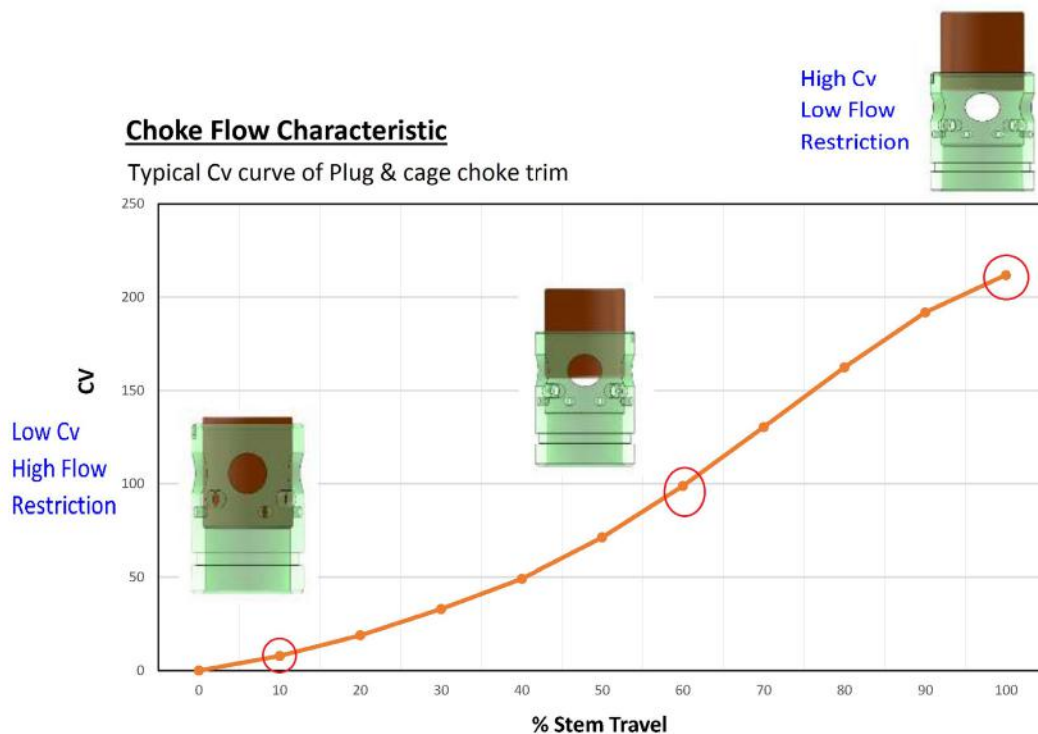
Choke flow control falls into three broad trim characteristics.

- Quick Opening
- Linear
- Equal percentage: Percentage of choke opening is equal to the percentage of change in the flow capacity. It is an exponential curve that has better flow controllability at low opening positions.



Choke Flow Characteristic

Typical Cv curve of Plug & cage choke trim



Drilling Chokes

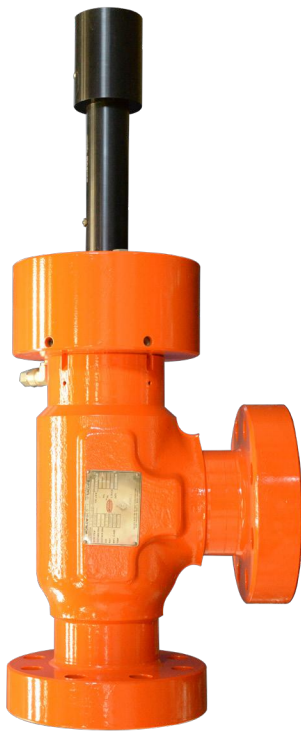
Manual and Hydraulic

WOM drilling chokes offer the customer increased service life and cost savings in high pressure drilling operations especially where substantial amount of abrasive fluids are present. WOM drilling chokes are designed as per API 16C.

WOM Drilling chokes are designed to encounter large cuttings coming up with the drilling mud. These chokes features cutting edge gate and seat design which reduces wear in extremely hostile flow conditions. Their consistent performance has been proved in kick control, well testing, well clean-up and other harshest of conditions. Available in 1-3/4" orifice trim with 5,000, 10,000 and 15,000 PSI pressure ratings.



WOM Manual Drilling Choke



WOM Hydraulic Drilling Choke

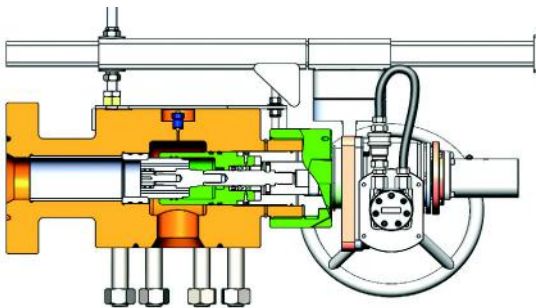
- Tungsten Carbide gate and seat have reversible design which increases life of the choke.
- Pressure balanced trim considerably reduces the torque required to operate the choke.
- Large body cavity around trim components reduces the speed of the solids in the fluids, thus enhances the body life.
- Extended wear sleeve limits downstream erosion to the trim, protecting the body from damage.
- Gate is fully guided by the Bonnet Extension which can be rotated with Bonnet to shift the surface wear.

Magnum Choke

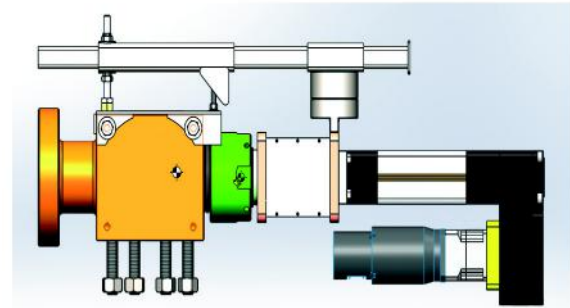
Magnum chokes are designed for extreme conditions like flow back, drilling, choke & kill, coil tubing, well control & snubbing.

The Magnum choke features a TC Seat & Gate as the throttling mechanism. The TC Gate moves up/down to vary choke opening area. The hydraulic motor operates the worm gear which in turn moves the TC Gate up & down to open and close the choke by overcoming the well fluid load in the choke body.

- Pressure balanced trim considerably reduces the torque required to operate the choke.
- Large body cavity around trim components reduces the velocity of the solids in the fluids, thus enhances the body life.
- Extended wear sleeve limits downstream erosion to the trim, protecting the body from damage.
- Gate is fully guided by the Bonnet Extension which can be rotated with Bonnet to shift the surface wear.
- The magnum choke can accommodate 3", 2" or 1.5" Max. Orifice trim with change in process requirements. All sizes of trims are interchangeable within the same choke body.
- Option for the automation upgrade (i.e hydraulic or electric)
- Standard Digital Position indicator -4-20 standard.
- Feedback Signal 4-20 MA available.



Hydraulic motor driven manual override



Linear Electric actuator driven (servo)

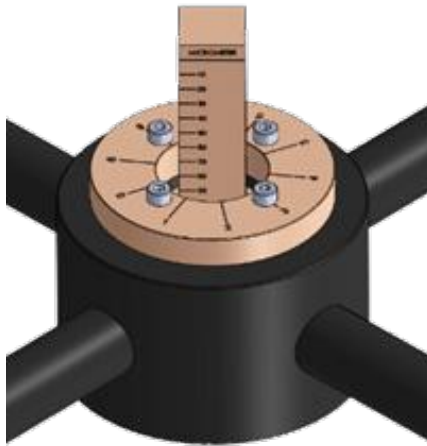
Available Sizes and Pressure Ratings Magnum Choke

NOMINAL SIZE	API PREASSURE RANGE (PSI)	MAX.ORIFICE SIZE	AVAILABLE TRIMS SIZE FRACTIONS
2"	2"	0.50"	32/64
3"	3"	0.75"	48/64
4" to 7"	4"	1.0"	64/64

Additional Features

Pressure Balancing Hole

Pressure balance hole is provided in all WOM choke valves so as to equalise the pressure around the Stem Seal which reduces the stem load & in turn the torque required to operate the valve.



Micrometer Indicator

Linear allowance in the Micrometer Indicator marking for easy calibration of the Indicator after assembly.

Large visual Dial Indicator for easy operation.

Stem Travel, % Open and Bean Size are shown on the Indicator Marking for easy & correct positioning of the valve.

Thumb Screw

Thumb Screw maintains set position of the Stem by resisting rotation of the drive nut.



Stem Packing & LIP Seals

Spring energized lip seals with scrapers used for dynamic seals enhance reliability of stem packing and pressure balance sealing.

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